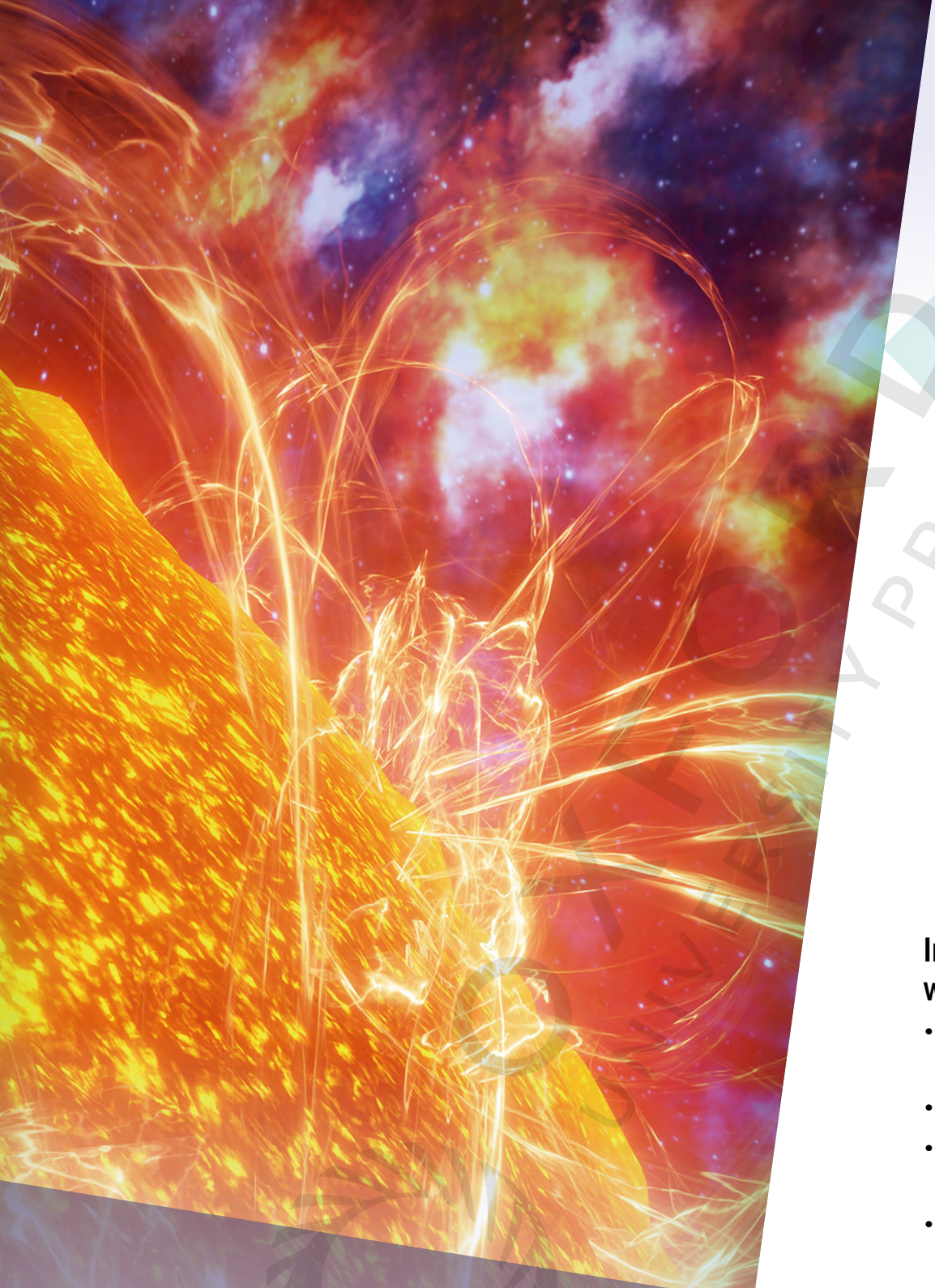


3

Nuclear Energy

In this chapter, we will learn about

- nuclear fission and chain reaction
- nuclear fusion
- the relationship between mass and energy
- the use of nuclear energy

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3.1 Nuclear fission and fusion

Nuclear reactions involve a change in the structure of nuclei. The radioactive decay is a kind of nuclear reaction. In this chapter, we will study two other kinds of nuclear reactions, namely **nuclear fission** and **nuclear fusion**.

Flipped classroom
Watch a video and answer the questions.

DSE goal
Realize the release of energy in nuclear fission.
Realize nuclear chain reaction.

DSE exam
18(1A)Q31, 19(1B)Q10, 22(1A)Q33, 23(1A)Q33

Naturally occurring uranium mainly consists of two isotopes, U-238 (99.27%) and U-235 (0.72%).

Not all nuclei can be fissioned with slow-moving neutrons, e.g. U-238 can only be fissioned with fast-moving ones. It is usually used to produce Pu-239 by absorbing a slow-moving neutron. Pu-239 is used as nuclear fuel in nuclear reactors and useful in producing nuclear weapons.

1 Nuclear fission

a Introduction to nuclear fission

In nuclear fission, a nucleus splits into several lighter nuclei. Neutrons may be emitted during the process. Nuclear fission can happen spontaneously or when a nucleus is bombarded by a neutron. For example, uranium-235 can undergo fission when it is bombarded by a slow-moving neutron (Fig 3.1a).

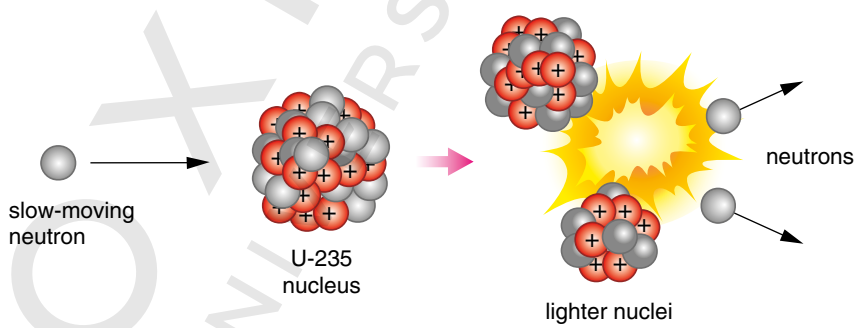


Fig 3.1a *An example of nuclear fission.*

A uranium-235 nucleus can split in many ways, for example:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{144}_{56}\mathrm{Ba} + {}^{90}_{36}\mathrm{Kr} + 2{}^{1}_{0}\mathrm{n} + \text{energy}$$
$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{143}_{54}\mathrm{Xe} + {}^{90}_{38}\mathrm{Sr} + 3{}^{1}_{0}\mathrm{n} + \text{energy}$$

If a heavy nucleus (e.g. uranium-235) undergoes nuclear fission, a huge amount of energy will be released. The energy is released in the form of electromagnetic radiation and the kinetic energy of the fission products.

Nuclear fission is a nuclear reaction in which a nucleus splits into lighter nuclei.

Know more

Discovery of nuclear fission
Several scientists made contributions to the discovery of nuclear fission. Do you know their stories?
▸ p.120

nuclear fission 核裂變　　nuclear fusion 核聚變

b Sustainable chain reaction and critical mass

Have you ever tried building up a line of dominoes? When you hit the first domino lightly, the remaining dominoes will topple one by one—the toppling of one domino triggers the toppling of the next (Fig 3.1b).

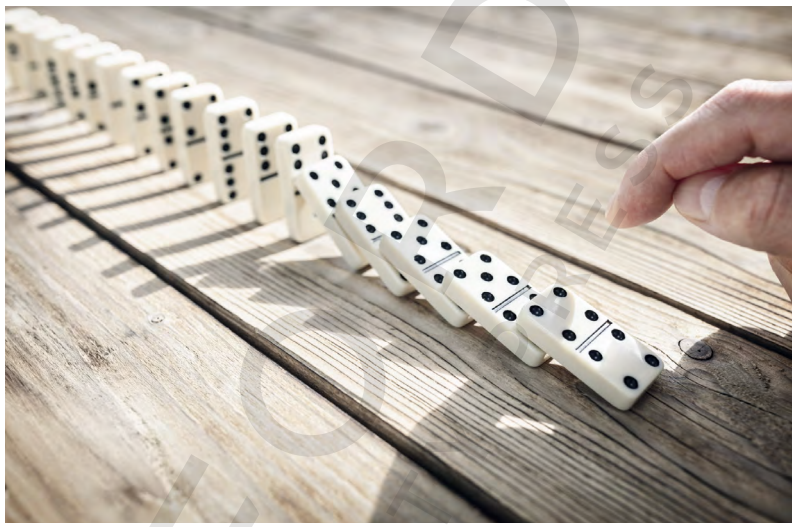


Fig 3.1b *A line of dominoes.*

Simulation

→ This simulation shows graphically how nuclear fission and chain reaction occur when uranium-235 nuclei are bombarded by neutrons, releasing a huge amount of energy in the process.

Nuclear fission can happen in a similar way. Under suitable conditions, the fission of one nucleus can set off a series of fission reactions. For example, when a uranium-235 nucleus undergoes fission, neutrons are emitted. Some of these neutrons may escape while some may split other uranium-235 nuclei and emit more neutrons. This triggers further fission reactions and results in a **chain reaction** (Fig 3.1c).

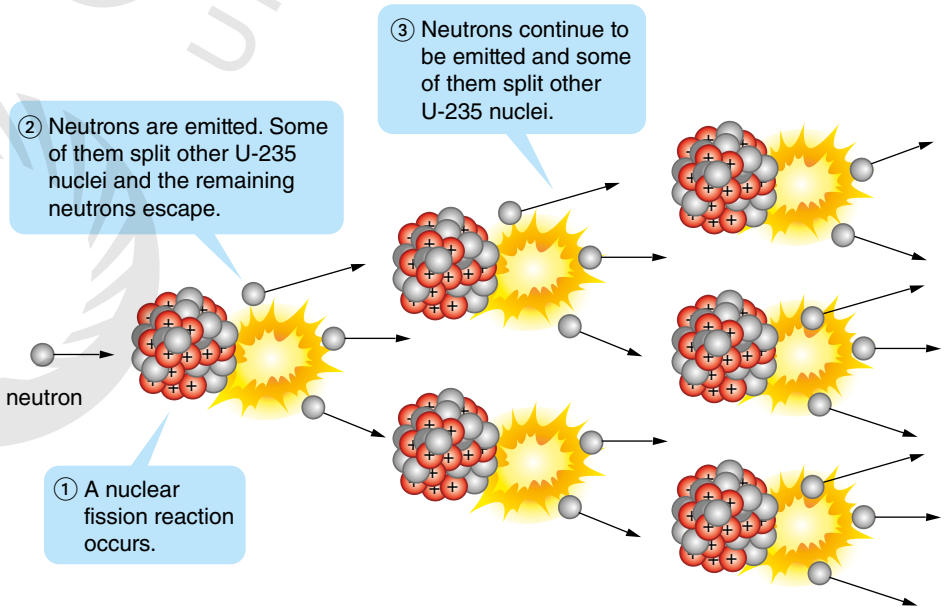


Fig 3.1c *A chain reaction of uranium-235.*

chain reaction 連鎖反應

However, a chain reaction will not last indefinitely. If fewer than one neutron on average from each fission causes another fission, the chain reaction will eventually stop. To sustain a chain reaction, a sufficient number or mass of *fissile* nuclei must be present. The minimum amount of fissile material required for a sustained chain reaction is called the **critical mass** (Fig 3.1d).

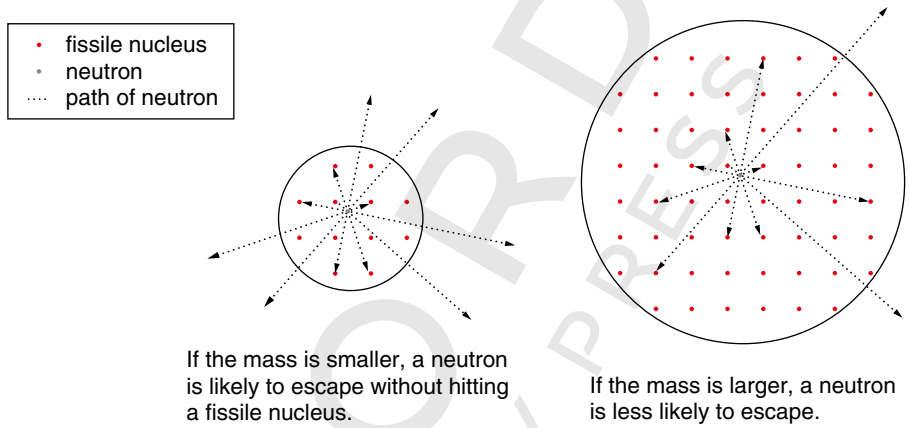


Fig 3.1d *A chain reaction requires a sufficient mass of fissile material.*

A chain reaction is sustained if at least one neutron on average from each fission causes another fission. The minimum amount of fissile material required for a sustained chain reaction is called the critical mass.

Example 1 | Fission of plutonium

A plutonium-239 ($^{239}_{94}$Pu) nucleus is bombarded by a neutron and undergoes fission. It splits into a molybdenum-104 ($^{104}_{42}$Mo) nucleus and a tellurium-134 ($^{134}_{52}$Te) nucleus, releasing x neutrons.

(a) Write an equation for the reaction in terms of x. Hence, find x.

(b) If there is enough Pu-239 fuel, is it possible for this reaction to trigger a chain reaction? Explain.

Solution

(a) $^{239}_{94}\text{Pu} + {}^{1}_{0}\text{n} \rightarrow {}^{104}_{42}\text{Mo} + {}^{134}_{52}\text{Te} + x\,{}^{1}_{0}\text{n}$

Consider the mass numbers.

$239 + 1 = 104 + 134 + x \times 1$

$x = 2$

(b) Yes, because more than one neutron is produced in each fission, which can trigger further fissions.

fissile 可裂變的　　critical mass 臨界質量

2 Nuclear fusion

In nuclear fusion, two nuclei join together to form a heavier nucleus and other particles. For example, a hydrogen-2 nucleus and a hydrogen-3 nucleus can fuse to form a helium-4 nucleus and a neutron (Fig 3.1e).

Hydrogen-2 is called deuterium and hydrogen-3 is called tritium.

DSE goal

Realize the release of energy in nuclear fusion.

Realize nuclear fusion as the source of solar energy.

DSE exam

15(1A)Q31, 15(1B)Q10(b), 22(1B)Q10(b), (c)

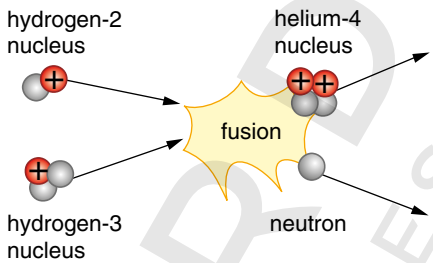


Fig 3.1e An example of nuclear fusion.

Simulation

→ This simulation shows graphically the fusion of a hydrogen-2 nucleus and a hydrogen-3 nucleus. The fusion produces a helium-4 nucleus and a neutron, and releases a huge amount of energy.

The above fusion reaction can be written as:

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + \text{energy}$$

If two light nuclei (e.g. hydrogen-2 and hydrogen-3) fuse, a huge amount of energy will be released in the form of electromagnetic radiation as well as the kinetic energy of the fusion products.

Nuclear fusion cannot be triggered easily because nuclei carry positive charges (protons) and repel each other. Two nuclei can fuse only when they have very high kinetic energy—this allows them to overcome the electrostatic repulsion and get very close to each other. The temperature required to start a fusion reaction is typically about 10^7 K.

Recall from Book 1 Ch 5 that temperature is a measure of the average KE of the molecules in a gas.

Nuclear fusion takes place in the Sun and other stars. In the core of the Sun, hydrogen nuclei fuse into helium nuclei, resulting in a release of energy that keeps the Sun hot (Fig 3.1f). In other words, nuclear fusion is the source of solar energy.

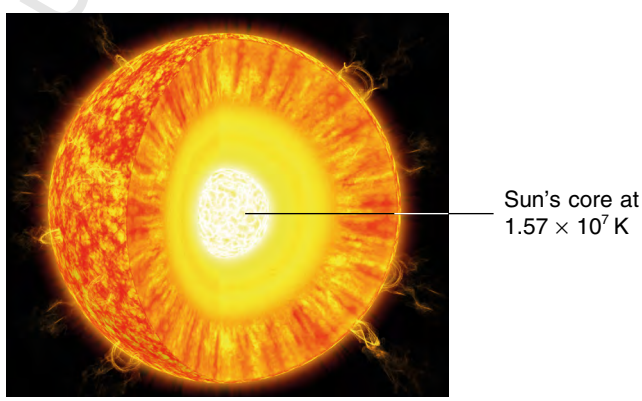


Fig 3.1f Nuclear fusion of hydrogen nuclei takes place in the Sun's core.

In nuclear fusion, two nuclei join together to form a heavier nucleus. A high temperature is required to start a fusion reaction.

Table 3.1a summarizes what we learned about radioactive decay, nuclear fission and nuclear fusion.

	Radioactive decay	Nuclear fission	Nuclear fusion
Nature	an unstable nucleus releases energy through nuclear radiation	a nucleus splits into lighter nuclei	two nuclei join together to form a heavier nucleus
Feasibility	spontaneous (no trigger)	can happen spontaneously or when bombarded by a neutron	not spontaneous and takes place at a very high temperature
Energy released	low	high	high
Example	$^{60}_{27}\text{Co} \rightarrow {}^{60}_{28}\text{Ni} + {}^{0}_{-1}\text{e}$ 0.007 pJ of energy is released per nucleon	$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{144}_{56}\text{Ba} + {}^{90}_{36}\text{Kr} + 2{}^{1}_{0}\text{n}$ 0.122 pJ of energy is released per nucleon	$^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n}$ 0.564 pJ of energy is released per nucleon

$1\ \text{pJ} = 10^{-12}\text{J}$

The energy released per nucleon is equal to the total energy released divided by the total number of nucleons of the starting elements.

Usually, nuclear fusion releases the largest amount of energy per nucleon. Therefore, nuclear fusion releases the largest amount of energy with the same mass of nuclear fuel. ▶

Table 3.1a *Summary of radioactive decay, nuclear fission and nuclear fusion.*

☆ HKDSE 2015 Paper 1A Q31

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Spontaneous reaction

Which of the following is/are spontaneous reaction(s)?

(1) $^{239}_{94}\text{Pu} + {}^{1}_{0}\text{n} \rightarrow {}^{134}_{54}\text{Xe} + {}^{103}_{40}\text{Zr} + 3{}^{1}_{0}\text{n}$

(2) $^{239}_{94}\text{Pu} \rightarrow {}^{235}_{92}\text{U} + {}^{4}_{2}\text{He}$

(3) $^{15}_{7}\text{N} + {}^{1}_{1}\text{H} \rightarrow {}^{12}_{6}\text{C} + {}^{4}_{2}\text{He}$

A (2) only
B (3) only
C (1) and (2) only
D (2) and (3) only

Solution

(1) and (3) are both reactions resulting from the interaction of two particles and therefore are not spontaneous. On the other hand, (2) does not involve the interaction of more than one particle.

∴ (1), (3) are incorrect and (2) is correct.

∴ The answer is A.

Common mistake ⚠
Students may not realize that a reaction is not spontaneous as long as there is more than one reacting particle.

▶ Revision 3 Q8 (p.116)

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The nuclear reaction that a nucleus splits into two or more lighter nuclei is called (nuclear fission / nuclear fusion / chain reaction).

2 The nuclear reaction that two nuclei combine to form a heavier nucleus is called (nuclear fission / nuclear fusion / chain reaction).

3 The fission of uranium-235 is triggered by

A a slow-moving electron.
B a slow-moving neutron.
C a fast-moving electron.
D a fast-moving neutron.

4 Energy is released in the fission of

A light nuclei.
B heavy nuclei.
C any nuclei.

5 Energy is released in the fusion of

A light nuclei.
B heavy nuclei.
C any nuclei.

6 For a chain reaction of fission of U-235 to be sustained,

A no neutron should be produced in each fission.
B only one neutron should be produced in each fission and it bombards another U-235 nucleus to trigger another fission.
C more than one neutron is produced in each fission and at least one bombards another U-235 nucleus to trigger another fission.
D more than one neutron is produced in each fission and all of them bombard other U-235 nuclei to trigger fissions.

7 Critical mass is the (minimum / maximum) amount of (fissile material / neutrons) that can sustain a chain reaction. If the amount is less than the critical mass, there will be too many (fissile nuclei / neutrons) escaping without taking part in any further reaction.

Exercise 3.1

1 A large amount of energy is necessary to start a nuclear fusion reaction because

A a chain reaction has to be sustained.
B electrostatic repulsion exists between nuclei.
C a significant amount of energy is produced in nuclear fusion.
D energy is needed to melt the nuclei.

★**2** Consider the following nuclear reaction.

$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{131}_{53}\mathrm{I} + {}^{102}_{p}\mathrm{Y} + q\,{}^{1}_{0}\mathrm{n}$

(a) What type of nuclear reaction does it belong to?
(b) Find the values of p and q.
(c) If there is enough U-235 fuel, is it possible for this reaction to trigger a chain reaction? Explain.

★**3** Consider the following nuclear reaction.

$^{4}_{2}\mathrm{He} + {}^{x}_{2}\mathrm{He} \rightarrow {}^{7}_{y}\mathrm{Be} + \text{energy}$

(a) What type of nuclear reaction does it belong to?
(b) Find the values of x and y.

★**4** A plutonium-239 ($^{239}_{94}$Pu) nucleus is bombarded by a neutron and splits into a xenon-134 ($^{134}_{54}$Xe) nucleus and a zirconium-103 ($^{103}_{40}$Zr) nucleus.

(a) Write down an equation for this reaction.
(b) If there is enough Pu-239 fuel, is it possible for this reaction to trigger a chain reaction? Explain.

★**5** Two carbon-12 ($^{12}_{6}$C) nuclei undergo fusion to produce a magnesium-23 ($^{23}_{12}$Mg) nucleus and another particle.

(a) Write down an equation for this fusion reaction.
(b) Explain why a high temperature is required for this reaction to occur.

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Ext

3.2 Mass-energy relationship

1 Mass and energy

DSE goal

State mass-energy relationship $\Delta E = \Delta mc^2$.

DSE exam

13(1A)Q36

The Sun (Fig 3.2a) is fuelled by hydrogen which undergoes nuclear reactions in the Sun's core. Every second the Sun releases about 3.85×10^{26} J of energy! Where does such a large amount of energy come from? The mass of hydrogen!

Mass can be converted into energy; energy can also be converted into mass. When mass is converted into energy, the amount of energy released is given by the **mass-energy relationship**:

energy released (J) ····· loss in mass (kg)

$$\Delta E = \Delta mc^2$$

····· speed of light $\approx 3.00 \times 10^8$ m s^{-1}

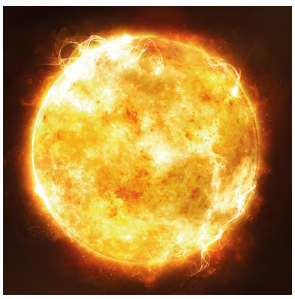


Fig 3.2a *The Sun is releasing energy.*

Since c is a large number, a huge amount of energy is released from a tiny loss in mass. 1 kg of mass is equivalent to 9×10^{16} J of energy.

Example 2 Mass-energy relationship

The amount of energy released by the Sun in each second is 3.85×10^{26} J and the energy is converted from the mass of the Sun. Find the mass lost by the Sun every month. Take 1 month = 2.59×10^6 s.

Solution

Consider a period of 1 month.

Energy released $\Delta E = (3.85 \times 10^{26}) \times (2.59 \times 10^6) = 9.97 \times 10^{32}$ J

By $\Delta E = \Delta mc^2$,

$$\Delta m = \frac{\Delta E}{c^2} = \frac{9.97 \times 10^{32}}{(3.00 \times 10^8)^2} = 1.11 \times 10^{16} \text{ kg}$$

The Sun loses 1.11×10^{16} kg every month.

Know more

Albert Einstein

The mass-energy relationship was discovered by the great scientist Albert Einstein. He proposed the *special theory of relativity* in 1905 and the mass-energy relationship is one of the remarkable results of the theory. This result was very important in the study of nuclear reactions.

▸ p.120

Later, we will learn how to work out the energy released in a nuclear reaction by considering the change in the total mass of particles. Before that, we need to learn about some new units of energy and mass.

mass-energy relationship 質能關係　　special theory of relativity 狹義相對論

Ext

DSE goal
Use atomic mass unit as a unit of energy.

2 Units of energy and mass in the atomic scale

When a single nuclear reaction is considered, the SI units joule (J) and kilogram (kg) are too large. Therefore, scientists introduce two units which are more convenient to express energy and mass in the atomic scale.

a Electron-volt

Cross-link
You learned about electric force in Book 4 Ch 1.

In Figure 3.2b, a positive charge q is placed in a uniform electric field produced by two parallel charged plates A and B. It experiences an electric force. When q is released, it will accelerate towards the negative plate B and gain KE.

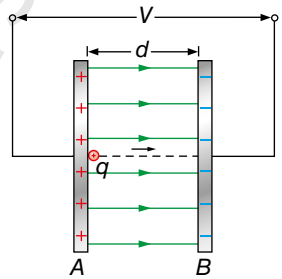


Fig 3.2b Charge q accelerating across two parallel charged plates.

Consider the motion of q from A to B. The kinetic energy E gained by q is given by

Electric PE is converted into KE as work is done by electric force.
Refer to Book 4 p.54 for the introduction to electric potential energy.

$$E = Fs = \left(\frac{Vq}{d}\right)(d) = qV$$

where V is the potential difference across A and B.

If q is a negative charge, it will still accelerate in the electric field but in the opposite direction. The amount of KE gained by q as it moves from B to A is worked out by the same equation.

Consider an electron being accelerated across two points of a potential difference 1 V. The amount of KE gained by the electron is said to be one **electron-volt** (**eV**) (Fig 3.2c). The eV is often used as a unit of energy in the atomic scale.

The conversion between eV and J is as follows:

$$\begin{aligned} 1 \text{ eV} &= e \times 1 \text{ V} \\ &= (1.60 \times 10^{-19} \text{ C}) \times (1 \text{ V}) \\ &= 1.60 \times 10^{-19} \text{ J} \end{aligned}$$

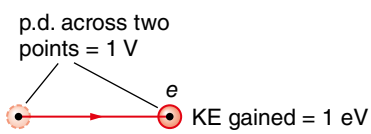


Fig 3.2c The definition of 1 eV.

When a larger amount of energy is measured, we may use MeV as a unit, where 1 MeV = 10^6 eV.

One electron-volt is the KE gained by an electron when it is accelerated across two points of a potential difference 1 V.

$$\mathbf{1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}}$$

electron-volt (eV) 電子伏

Ext

b Atomic mass unit

The **atomic mass unit** (**u**) is a common unit of mass in the atomic scale. It is defined as (Fig 3.2d):

1 u is about the mass of a nucleon.

$$\textbf{1 u} = \frac{1}{12} \times \textbf{mass of a carbon-12 atom} = \textbf{1.661} \times \textbf{10}^{-27}\ \textbf{kg}$$

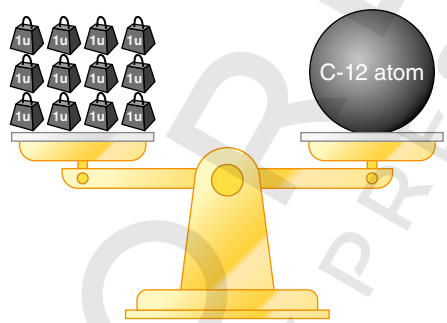


Fig 3.2d Atomic mass unit

Ts may remind Ss that the mass of an atom is different from that of its nucleus. The former includes the mass of the electrons as well. For example, the mass of a He-4 atom is 4.002 603 u.

Table 3.2a shows the masses of some particles in atomic mass unit.

Particle	Mass / u
Proton	1.007 276
Neutron	1.008 665
Electron	0.000 549
Hydrogen-2 (deuterium) nucleus	2.013 553
Helium-4 nucleus	4.001 505
Polonium-210 nucleus	209.936 758
Uranium-235 nucleus	234.993 422

Table 3.2a The masses of some particles in atomic mass unit.

As mass and energy are equivalent, the atomic mass unit is often used as a unit of energy.

$\Delta E = \Delta mc^2$

Energy equivalent to 1 u of mass $= (1.661 \times 10^{-27}) \times (3.00 \times 10^{8})^2$ J

$= 1.49 \times 10^{-10}$ J

$= \dfrac{1.49 \times 10^{-10}}{1.60 \times 10^{-19}}$ eV

$= 9.31 \times 10^{8}$ eV

$= 931$ MeV

1 u = 1.661 × 10^{-27} kg
1 u is equivalent to 931 MeV.

atomic mass unit (u) 原子質量單位

Ext

→ Teaching notes and exercises are available on OUP web.

Skill
Problem-solving

Conversion between mass and energy in different units

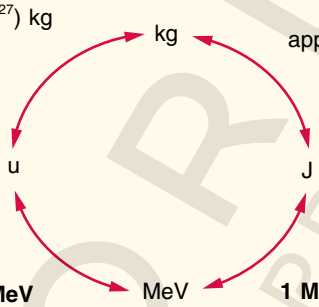
The figure below helps us solve problems when making conversions between mass and energy.

1 u = 1.661 × 10^{-27} kg

p u = $p \times (1.661 \times 10^{-27})$ kg

q kg = $\dfrac{q}{1.661 \times 10^{-27}}$ u

p and q are numbers with no unit. ▶



apply $\Delta E = \Delta mc^2$

1 u is equivalent to 931 MeV

p u is equivalent to $p \times 931$ MeV

q MeV is equivalent to $\dfrac{q}{931}$ u

1 MeV = 1 × 10^6 × (1.60 × 10^{-19}) J

p MeV = $p \times 10^6 \times (1.60 \times 10^{-19})$ J

q J = $\dfrac{q}{10^6 \times (1.60 \times 10^{-19})}$ MeV

Example 3 | Units of mass and energy

The mass of a uranium-235 nucleus is 234.993 422 u.

(a) Express the mass in kg.

(b) Hence find the amount of energy equivalent to this mass. Express your answer in J.

(c) Find the amount of energy equivalent to 1 kg of uranium-235. Express your answer in MeV.

Solution

(a) Mass = $234.993\ 422 \times (1.661 \times 10^{-27})$
$= 3.90 \times 10^{-25}$ kg

(b) Energy = $(3.90 \times 10^{-25}) \times (3.00 \times 10^8)^2$
$= 3.51 \times 10^{-8}$ J

Alternative method:
Energy = $(1) \times (3.00 \times 10^8)^2$
$= 9.00 \times 10^{16}$ J
$= \dfrac{9.00 \times 10^{16}}{1.60 \times 10^{-19}}$ eV
$= 5.63 \times 10^{35}$ eV
$= 5.63 \times 10^{29}$ MeV

(c) Energy = $1 \times \dfrac{1}{1.661 \times 10^{-27}} \times 931$
$= 5.61 \times 10^{29}$ MeV

Tip

$1 \times \dfrac{1}{1.661 \times 10^{-27}}$ gives the mass in u. ▶

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Ext

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 *True or false*: If the loss in mass during a nuclear reaction is 0.01 u, the energy equivalent to this mass is $0.01 \times (3 \times 10^8)^2$ J. (T/F)

2 Complete the following table.

	Particle	Mass / kg	Mass / u
(a)	Carbon-12 atom		12.0000
(b)	Carbon-14 atom	2.3259×10^{-26}	
(c)	Uranium-238 nucleus	3.9532×10^{-25}	
(d)	Roentgenium-272 nucleus		272.0923

3 Complete the following table.

Answers obtained using different methods may vary slightly.

	Loss in mass / u	Loss in mass / kg	Energy released / J	Energy released / MeV
(a)		1.44×10^{-27}		
(b)	0.193			
(c)			8.50×10^{-12}	
(d)				5.60

Learn by practice 1

1 A radioactive source emits 180 J of energy. Calculate the mass (in kg) equivalent to this amount of energy.

2 Two particles, each of mass 9.11×10^{-31} kg, collide into each other and are completely converted into energy. Find this energy in MeV.

3 The mass of a water molecule is 18.010 565 u. Express its mass in kg.

4 The mass lost in a nuclear reaction is 0.002 u. Find the equivalent energy in MeV and J.

DSE goal

Determine the energy release in nuclear reactions.

Apply $\Delta E = \Delta mc^2$ to solve problems.

DSE exam

12(1B)Q11(b), 15(1B)Q10, 17(1A)Q33, 18(1A)Q33, 19(1A)Q33, 20(1B)Q10, 22(1B)Q10(a), 23(1A)Q32, etc.

3 Energy released in nuclear rections

Recall that energy is released in **the fission of heavy nuclei and the fusion of light nuclei**. In these reactions, the total mass of the particles after the reaction is slightly less than before (Fig 3.2e on p.103). The **loss in mass** is converted into energy and released in the form of electromagnetic radiation as well as the kinetic energy of the fission products.

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Ext

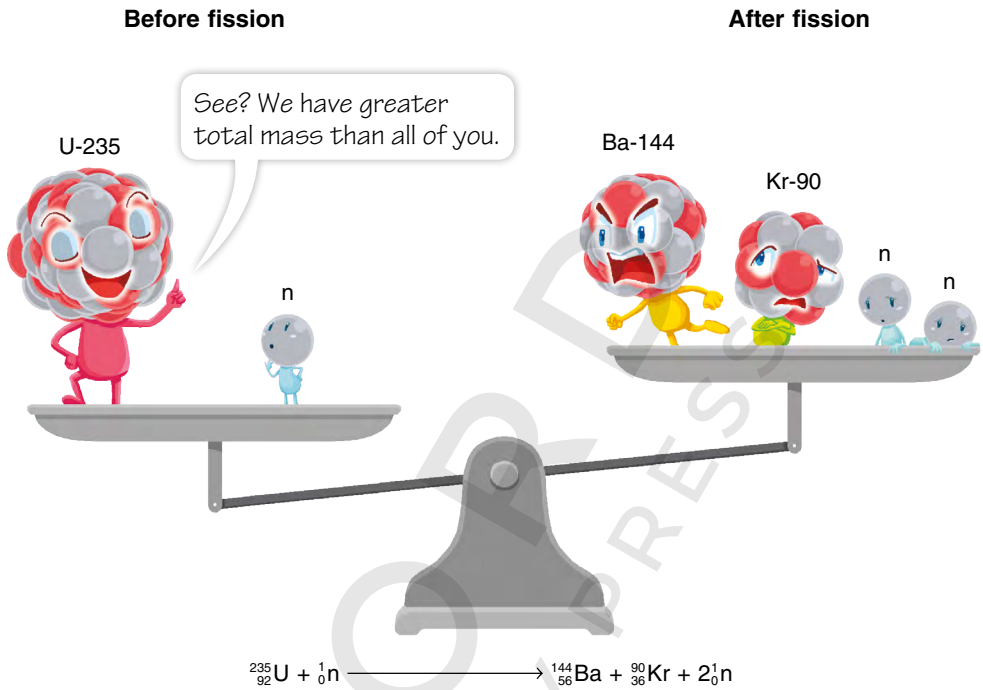


Fig 3.2e *Mass is lost when energy is released.*

The loss in mass also occurs in radioactive decays. Since the energy released in decay is much less than that released in nuclear fission and fusion, the loss in mass in decay is also much less.

The following examples show how the amount of energy is calculated in different nuclear reactions.

Example 4 Energy released in nuclear fission

Consider the following nuclear fission reaction.

$$\underbrace{{}^{235}_{92}U + {}^{1}_{0}n}_{236.002\ 087\ u} \rightarrow \underbrace{{}^{144}_{56}Ba + {}^{90}_{36}Kr + 2{}^{1}_{0}n}_{235.809\ 305\ u}$$

(a) Find the loss in mass in the reaction. Express your answer in u.

(b) Hence find the energy released in the reaction. Express your answer in J.

Solution

(a) $\Delta m = 236.002\ 087 - 235.809\ 305$

$= 0.192\ 782$ u

(b) $\Delta E = \Delta mc^2 = (0.192\ 782 \times 1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$

$= 2.88 \times 10^{-11}$ J

Tip

Convert the unit of Δm from u to kg before applying $\Delta E = \Delta mc^2$.

Ext

Example 5 Energy released in nuclear fusion

Consider the nuclear fusion reaction below:

$${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$$

The table shows the masses of these particles.

Particle	Mass / u
Neutron	1.008 665
Hydrogen-2 nucleus	2.013 553
Hydrogen-3 nucleus	3.015 501
Helium-4 nucleus	4.001 505

(a) Find the loss in mass in the reaction. Express your answer in u.

(b) Hence find the energy released in the reaction. Express your answer in MeV.

Solution

(a) $\Delta m = (2.013\ 553 + 3.015\ 501) - (4.001\ 505 + 1.008\ 665) = 0.018\ 884$ u

(b) $\Delta E = 0.018\ 884 \times 931 = 17.6$ MeV

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 *True or false*:

(a) In the nuclear fission reaction ${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{143}_{54}\mathrm{Xe} + {}^{90}_{38}\mathrm{Sr} + 3{}^{1}_{0}\mathrm{n}$, the mass of $({}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n})$ is greater than that of $({}^{143}_{54}\mathrm{Xe} + {}^{90}_{38}\mathrm{Sr} + 3{}^{1}_{0}\mathrm{n})$. (T/F)

(b) In the nuclear fusion reaction ${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}$, the mass of $({}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H})$ is smaller than that of $({}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n})$. (T/F)

2 In the fission of a uranium-235 nucleus, the mass of the nucleus

A is completely converted into energy.

B is partly converted into energy.

C is unchanged.

D has increased.

3 Consider the following nuclear reaction.

$${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{138}_{55}\mathrm{Cs} + {}^{96}_{37}\mathrm{Rb} + 2{}^{1}_{0}\mathrm{n}$$

234.993 422 u 1.008 665 u 137.880 822 u 95.913 820 u

(a) Find Δm in u.

(b) Find ΔE in J and MeV.

☆ HKDSE 2012 Paper 1B Q11, 2014 Paper 1B Q10

→ This video demonstrates how this question is solved.

Video

Ext

DSE practice 2 | Energy released in an α decay

Polonium-210 is a radionuclide used in some early experiments involving radioisotope thermoelectric generators. It undergoes α decay into lead-206 with a decay constant of 5.80×10^{-8} s^{-1}.

$$^{210}_{84}\text{Po} \rightarrow {}^{206}_{82}\text{Pb} + {}^{4}_{2}\text{He}$$

Given: mass of $^{210}_{84}$Po nucleus = 209.936 758 u
mass of $^{206}_{82}$Pb nucleus = 205.929 448 u
mass of $^{4}_{2}$He nucleus = 4.001 505 u
1 u = 1.661×10^{-27} kg
Avogadro's number $N_A = 6.02 \times 10^{23}$

(a) Find the energy released in the reaction. Express your answer in joules. (3 marks)

(b) The fuel in a generator contains 2.77 g of polonium-210.

(i) Find the number of polonium-210 nuclei, hence the number of moles, in the fuel. (3 marks)

(ii) Estimate the power delivered by the fuel. (3 marks)

Solution

(a) $\Delta m = 209.936\ 758 - (205.929\ 448 + 4.001\ 505)$ 1M
$= 0.005\ 805$ u

$\Delta E = \Delta m c^2$
$= (0.005\ 805 \times 1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ 1M
$= 8.68 \times 10^{-13}$ J 1A

Common mistake
Students may make mistakes in the conversion between u, MeV and J.

(b) (i) Number of polonium nuclei N

$$= \frac{2.77 \times 10^{-3}}{209.936\ 758 \times 1.661 \times 10^{-27}}$$ 1M
$= 7.94 \times 10^{21}$

Number of moles $n = \frac{N}{N_A}$

Common mistake
Students may make mistakes in the conversion between mass, number of particles and number of moles.

$= \frac{7.94 \times 10^{21}}{6.02 \times 10^{23}}$ 1M
$= 0.0132$ mol 1A

(ii) Activity of the fuel $A = kN$
$= (5.80 \times 10^{-8}) \times (7.94 \times 10^{21})$ 1M
$= 4.61 \times 10^{14}$ Bq

Power delivered
= energy released in each decay × number of decays in 1 s
$= (8.68 \times 10^{-13}) \times (4.61 \times 10^{14})$ 1M
= 400 W 1A

▶ Revision 3 Q15, 16 (p.118)

EXT

Exercise 3.2

Take 1 u = 1.661×10^{-27} kg, 1 u is equivalent to 931 MeV, $c = 3.00 \times 10^{8}$ m s^{-1}.

1 Which of the following is a correct definition of the atomic mass unit?

A Mass of a neutron

B Mass of a hydrogen-1 atom

C $\frac{1}{4}$ × mass of a helium-4 atom

D $\frac{1}{12}$ × mass of a carbon-12 atom

2 The Sun keeps generating energy by nuclear fusion. What can be deduced from this statement?

A The mass of the Sun keeps decreasing.

B The mass of the Sun keeps increasing.

C The size of the Sun keeps decreasing.

D The size of the Sun keeps increasing.

★**3** Arrange the following amounts of energy in ascending order.

(1) Energy equivalent to 10 u

(2) 2000 MeV

(3) 3×10^{-9} J

A (1), (2), (3)

B (2), (1), (3)

C (3), (1), (2)

D (3), (2), (1)

4 The mass of a chocolate is 13.2 g (Fig a).



Fig a

Find the amount of energy released if it were converted completely into energy.

★**5** The equation below shows a nuclear fission reaction.

$$^{239}_{94}\text{Pu} + ^{1}_{0}\text{n} \rightarrow ^{134}_{54}\text{Xe} + ^{103}_{40}\text{Zr} + 3^{1}_{0}\text{n}$$

Given: mass of $^{239}_{94}$Pu = 239.000 558 u

mass of $^{134}_{54}$Xe = 133.875 749 u

mass of $^{103}_{40}$Zr = 102.905 231 u

mass of $^{1}_{0}$n = 1.008 665 u

(a) Find the loss in mass in u.

(b) Find the energy released in the above reaction in MeV.

★**6** The equation below shows a nuclear fusion reaction.

$$^{2}_{1}\text{H} + ^{3}_{2}\text{He} \rightarrow ^{4}_{2}\text{He} + ^{1}_{1}\text{p}$$

Given: mass of $^{2}_{1}$H = 2.013 553 u

mass of $^{3}_{2}$He = 3.014 931 u

mass of $^{4}_{2}$He = 4.001 505 u

mass of $^{1}_{1}$p = 1.007 276 u

(a) Find the loss in mass in kg.

(b) Find the energy released in the above reaction in J.

★**7** The equation below shows the decay of carbon-14.

$$^{14}_{6}\text{C} \rightarrow ^{14}_{7}\text{N} + ^{0}_{-1}\text{e}$$

Given: mass of $^{14}_{6}$C = 13.999 948 u

mass of $^{14}_{7}$N = 13.999 231 u

mass of an electron = 0.000 549 u

Avogadro's number = 6.02×10^{23} mol^{-1}

(a) Find the energy released in the above decay in MeV.

(b) How much energy is released if 0.2 mole of carbon-14 nuclei undergoes the decay? Express your answer in J.

★**8** In the nuclear reaction shown below, 3.268 MeV of energy is released.

$$^{2}_{1}\text{H} + ^{2}_{1}\text{H} \rightarrow ^{3}_{2}\text{He} + ^{1}_{0}\text{n}$$

The mass of a neutron is 1.008 665 u and that of a hydrogen-2 nucleus is 2.013 553 u. Find the mass of a helium-3 nucleus.

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3.3 Nuclear power

Nuclear fission can release a huge amount of energy. Such energy can be used to generate electricity if the rate of the reaction can be controlled.

1 Controlled nuclear fission

Current nuclear power plants make use of **controlled nuclear fission** to generate electricity. Typically, uranium-235 is used as the fuel. A nuclear power plant usually contains several **nuclear reactors** in which fission takes place.

In a nuclear reactor, the concentration of uranium-235 in the fuel is kept at a suitable level. Moreover, the number of neutrons that trigger further fission is carefully controlled. These measures can ensure the rate of fission will not be too low or too high. As a result, chain reaction is sustained while the rate of fission is kept to a safe limit. The energy released from fission is used to generate electricity (Fig 3.3a).

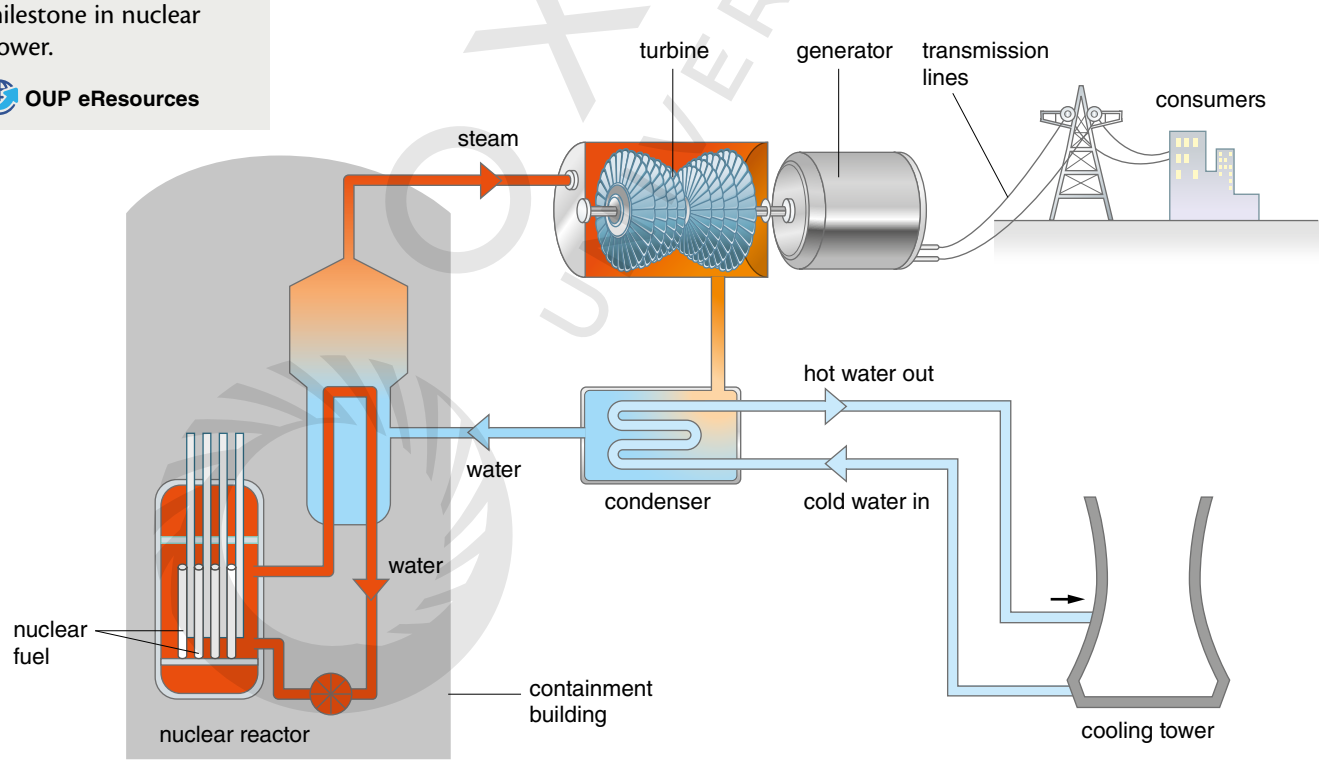


Fig 3.3a *Energy released from fission (in a nuclear reactor) is used to generate electricity.*

Know more

First controlled chain reaction

If a chain nuclear fission is not controlled, an enormous amount of energy will be released. This may result in an explosion. Conversely, a controlled chain reaction can produce a suitable amount of energy to generate electricity.

Enrico Fermi and his team first achieved a controlled chain reaction in 1942. This was a milestone in nuclear power.

OUP eResources

controlled nuclear fission 受控制裂變　　nuclear reactor 核反應堆　　Enrico Fermi 費米

2 Considerations regarding nuclear power NSE

Nuclear power offers an abundant supply of energy, but it also brings a lot of potential risks. If there is leakage of radioactive substances from **nuclear waste** disposal sites or from nuclear power plants, our health and the environment may be seriously affected. Table 3.3a shows some advantages and disadvantages of using nuclear power.

Surf the Internet to find out more advantages and disadvantages of nuclear power. What is your point of view about the use of nuclear power? Try to make your judgements scientifically and judiciously.

Advantages	Disadvantages
• Nuclear power can help relieve energy shortages. • Nuclear power's greenhouse gas emission rate is low compared to fossil fuels and many other energy sources. • The fuel costs are low compared to fossil fuels.	• Nuclear waste is produced. The waste is radioactive and usually contains radionuclides that have long half-lives. • Special handling and storage approaches for nuclear waste are needed (Fig 3.3b). • Nuclear accidents can be disastrous.

Table 3.3a *Some advantages and disadvantages of using nuclear power.*

For high-level waste, reusable materials, including unused uranium and a small quantity of plutonium-239, can be extracted from the waste in a reprocessing plant before sealing.

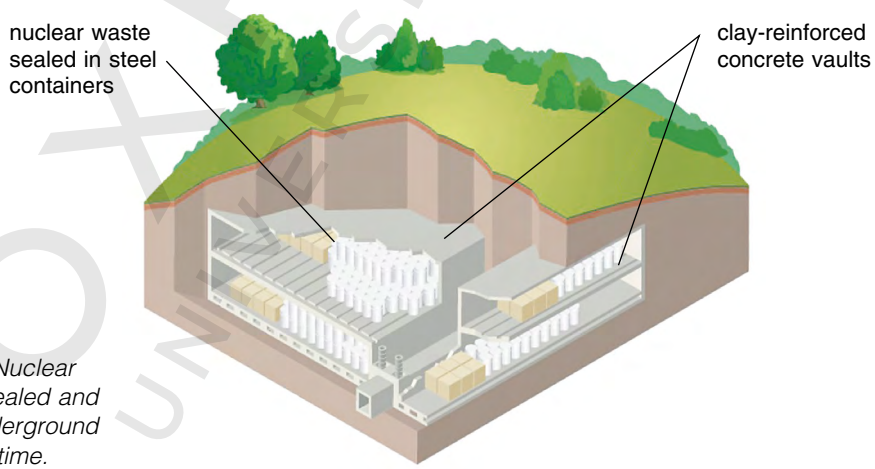


Fig 3.3b *Nuclear waste is sealed and stored underground for a long time.*

Carbon neutrality is a state of net-zero carbon dioxide emissions.

In order to achieve *carbon neutrality* in the future, a transition from fossil fuels to low-carbon energy sources is necessary. Nuclear power could be a recommendable energy source. In our country, nuclear power has been developing very quickly. It contributed to about 4.9% of the total electricity generation in 2019; it is believed that nuclear power will play an important role in replacing fossil fuels in the future.

Refer to 'Nuclear Safety' in CLP website for more detail.

Useful link

To minimize the potential risks, precautionary measures should be adopted in the design and operation of nuclear power plants. For example, the site of nuclear power plants should be carefully chosen to ensure the risks of natural hazards like flooding and severe storms are low. Moreover, the plants should be operated by qualified staff according to well-established safe practices and procedures.

nuclear waste 核廢料　　carbon neutrality 碳中和

Know more

Security considerations for the use of nuclear power NSE

While we enjoy the benefits brought by nuclear power, we should be aware of the importance of sustainable development. When developing nuclear power, effort should be made to minimize the impact on the environment.

- Ecological security is the basic requirement for life and should be safeguarded. The ecological system can be damaged by the construction and operation of nuclear power plants, the disposal of nuclear waste, etc. A meticulous plan should be devised to minimize the damage.
- Resource security is essential for national development. It includes the development, use and sustainable supply of resources like energy and water. The use of nuclear power should be aligned with the requirements for resource security.
- Nuclear security, which concerns nuclear fuels, nuclear facilities, nuclear technology etc., is important for humanity's future. When nuclear energy is developed, the risks of nuclear accidents, the spread of nuclear radiation, and other dangers should be minimized so that nuclear security can be safeguarded.

Useful link

Precautionary measures in nuclear power plants NSE

Guangdong Daya Bay Nuclear Power Station and *Lingao Nuclear Power Station* are situated near Hong Kong. Surf the Internet to see how these power plants implement the precautionary measures mentioned above. What other precautionary measures are taken in these facilities?

STSE

Radiation Monitoring and Assessment NSE

The Hong Kong Observatory (HKO) keeps track of the long-term changes in ambient radiation level in Hong Kong by using different advanced instruments. To monitor the effects of the nuclear power plants in Guandong, the HKO launched the *Environmental Radiation Monitoring Programme* in 1987. A range of methods are applied to extensively measure the amount of nuclear radiation and radionuclides present in the environment, including:

- Ground stations are set up to continuously measure the amount of radiation in fixed spots.
- Ground and aerial radiation surveys are regularly conducted.
- Samples from food, water, soil and the atmosphere are collected and the amount of radioactive substances in them is measured.

Besides the HKO, various government departments have been carrying out radiation monitoring and assessment work routinely, including:

- On-line Water Contamination Monitoring Systems operated by the Water Supplies Department
- Livestock Contamination Monitoring System run by the Food and Environmental Hygiene Department (FEHD)
- Food Radiation Monitoring run by FEHD at points of entry

Useful link

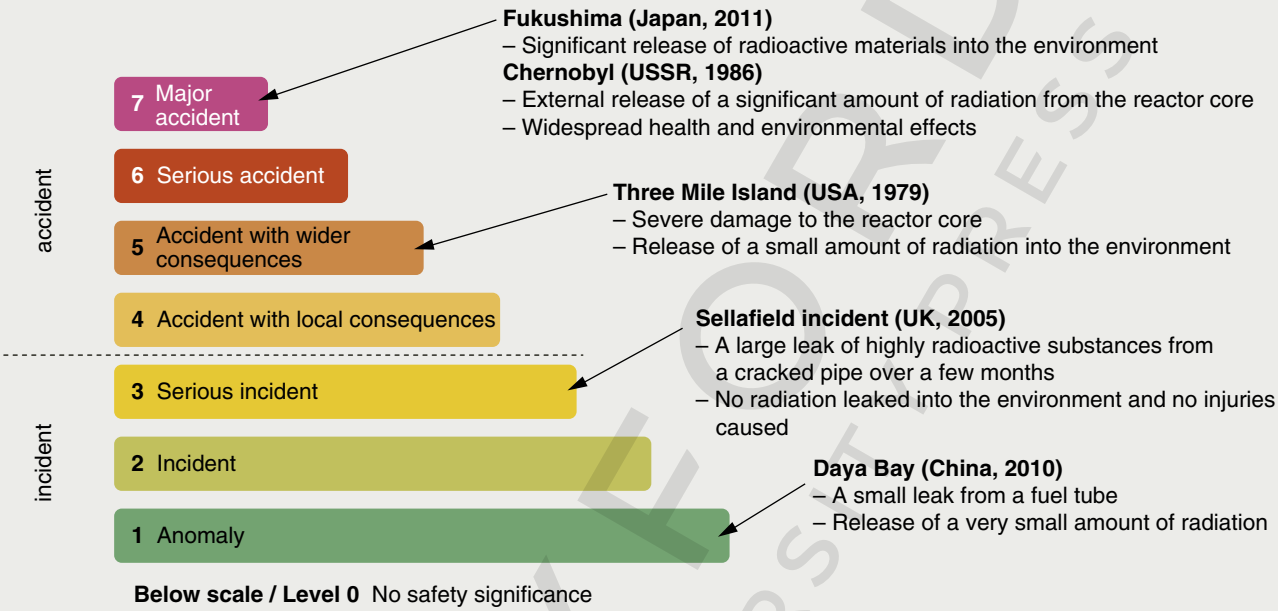
Guangdong Daya Bay Nuclear Power Station 廣東大亞灣核電站 Lingao Nuclear Power Station 嶺澳核電站
Environmental Radiation Monitoring Programme 環境輻射監測計劃

STSE

Nuclear accidents

International Nuclear and Radiological Event Scale

The International Nuclear and Radiological Event Scale (INES) classifies nuclear events into seven levels according to the impact of the events on three areas: the hazard caused on people and the environment outside the facility, the hazard caused within the facility and the damage to safety systems within the facility.



Major nucler accidents NSE

According to INES, there have been two major nuclear accidents so far. The latest one was the Fukushima nuclear accident that happened in 2011. An earthquake and an accompanying tsunami struck a nuclear power station in Fukushima. A series of explosions occurred and a large amount of radioactive substances leaked into the surrounding environment. About 140 000 residents were evacuated, and the soil, livestock and agricultural products in Fukushima were contaminated. Some people who participated in the rescue mission suffered from radiation sicknesses.



The Fukushima nuclear power plant was seriously damaged in the accident.

The other major accident was the Chernobyl nuclear accident. You may search for the details about the accident on the Internet.

Q16 Exam Report: Parts (a) and (b)(i) were well answered. It revealed that most candidates had a good understanding of the properties of α particles and the meaning of activity. In (b)(ii), only the more able ones knew that the power is given by the product of energy per decay and the activity found in (a)(i). Some wrongly used the total number of plutonium atoms in their calculation instead of the activity. In (b), for those who considered the number of half-lives elapsed in order to find the decrease in number of active atoms or activity, many obtained the correct numerical answer.

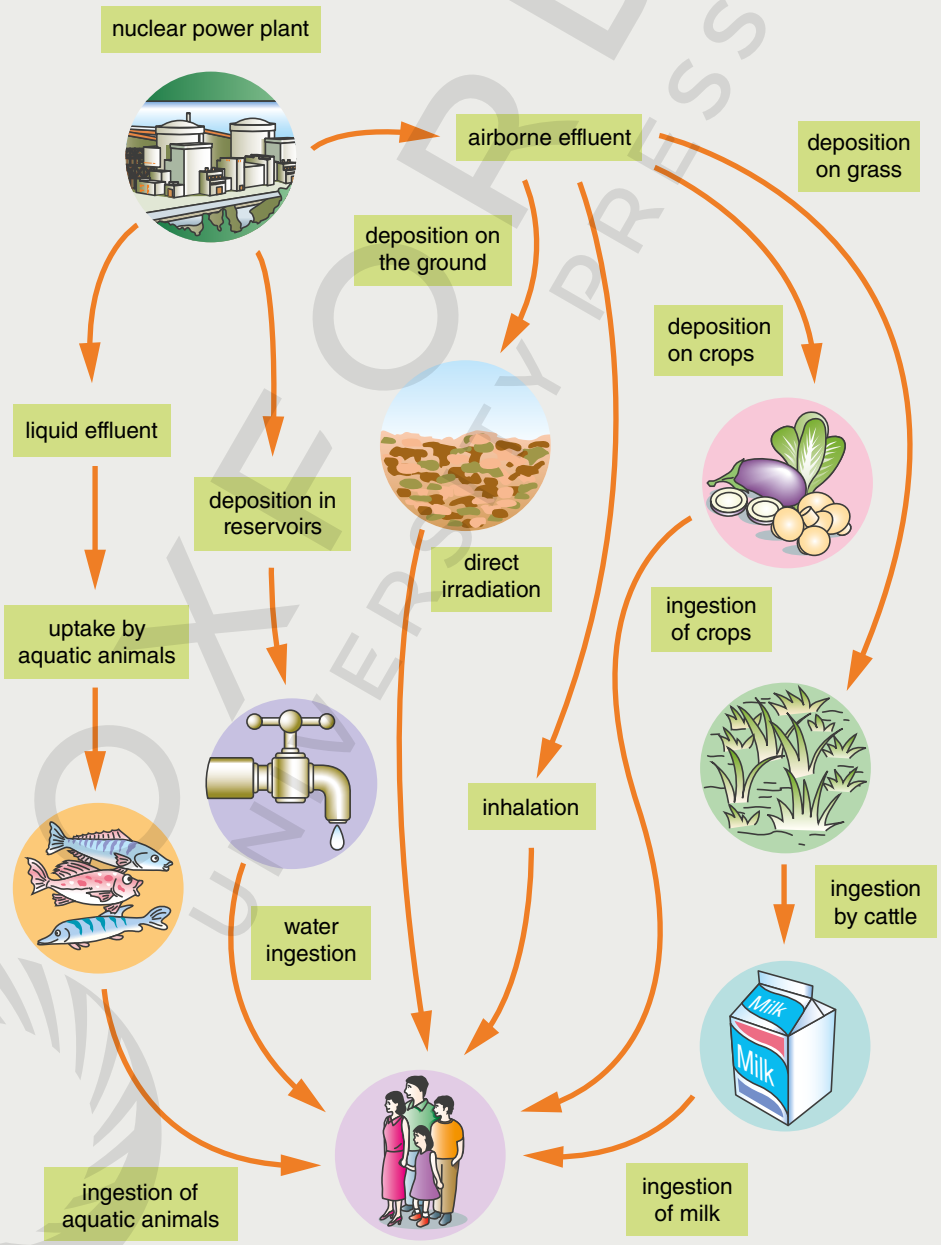
Q17 Exam Report: Candidates only did well in (a). A few candidates had difficulty in dealing with the units eV and J. In (b)(i), not many knew that the energy used for overcoming the repulsion between the two positive nuclei became electrical potential energy. Some wrongly thought that it turned into kinetic energy, heat or nuclear energy. In (b)(iii), most candidates failed to relate the average kinetic energy of the nuclei to the large amount of work done needed. A few even wrongly employed $E = 17.58$ MeV in (a) in their calculations.

Q18 Exam Report: Part (a)(i) was well answered. Not many gave the crucial condition that the number of neutrons produced in fission must be greater than one for a chain reaction be sustained in (a)(ii). Weaker candidates may have thought that slow neutrons had to be the products of the fission reaction. Most managed to find the amount of U-235 in the sample 2×10^9 years ago in (b)(i) but not many obtained the correct value of its concentration by mass in (b)(ii). Candidates' performance in (c) was poor. Very few were able to relate the energy of fission with the dry up of underground water, which led to a ceasing of the supply of slow neutrons. Wrong answers given by candidates included: the neutrons were not energetic enough or the concentration of the fuel was not high enough.

STSE

Nuclear safety NSE

Nuclear waste produced in power plants is sealed and buried underground. Under normal conditions, the radiation emitted by the waste can be shielded by the ground. Theoretically, the waste will not cause any effect on the environment except that it will heat up the ground. However, if nuclear waste leaks, radionuclides will pollute the environment and enter our bodies.



Radionuclides can be transferred to the environment and our bodies if nuclear waste leaks.

Following proper operating procedures, preventing accidents, and mitigating the consequences of accidents, as well as protecting the general public and the environment from radiation hazards are all important aspects of maintaining nuclear safety.

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3 Controlled nuclear fusion

Another way to generate nuclear power is to make use of **controlled nuclear fusion**. It is a safe source of energy. It is unlikely to cause nuclear accidents. Also, it does not produce any long-lived radioactive waste. Helium-4, the major product of fusion, is non-toxic and non-radioactive.

The supply of fuel, namely hydrogen-2 and hydrogen-3, is abundant. Hydrogen-2 can be obtained easily from the sea; hydrogen-3 can be artificially created. The amount of fuel used to produce the same amount of electricity is much less than that needed in a fission power plant and fossil fuel power plant.

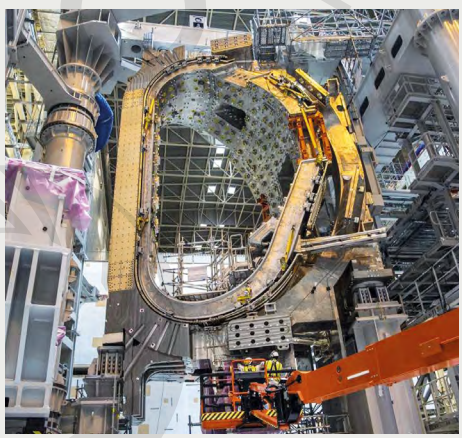
However, the technology required for controlled nuclear fusion is still under development. It is a challenging mission as a very high temperature (10^7 K) has to be maintained for fusion to occur. A fusion reactor which can withstand high temperatures is difficult to build.

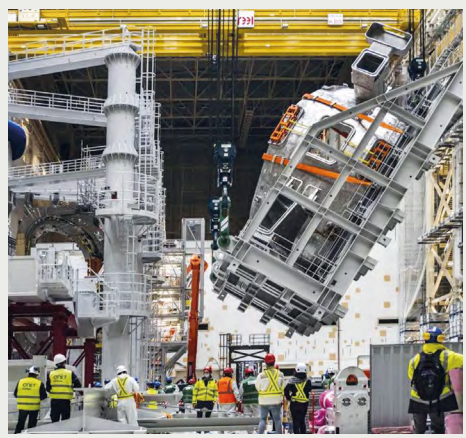
Know more

International Thermonuclear Experimental Reactor (ITER)

ITER, which began in the 1980s, is the world's largest research project for controlled nuclear fusion. It is a global collaboration of 35 nations. The major part of this project is to build the world's largest tokamak in France. The tokamak is an experimental machine designed to harness the energy of fusion. You may find out more about this project and the tokamak on the Internet.

Useful link





Transfer of sub-assemblys for the world's largest tokamak.

In May 2021, China maintained the temperature of the particles inside the tokamak fusion reactor at 120 million degree Celsius for 101 seconds. It is a project that China has been working on with the ITER. The development of fusion power may help meet China's energy requirements. It may also support sustainable development in China's energy sector and boost the national economy in the future.

controlled nuclear fusion 受控核聚變

STSE

Development of nuclear weapons

The development of nuclear technology has led to the invention of nuclear weapons. These weapons can rapidly release energy from **uncontrolled** nuclear fission or fusion reactions, thus causing massive destruction. Examples of nuclear weapons include the *atomic bomb* and the *hydrogen bomb*.

In an atomic bomb, highly enriched nuclear fuels are brought together to exceed its critical mass. A single atomic bomb is powerful enough to destroy an entire city. A hydrogen bomb consists of a fission bomb and a fusion fuel. When the fission bomb explodes, uncontrolled nuclear fusion reactions will be triggered in the fusion fuel.

Should nuclear weapons be developed?

The first nuclear weapons were built during World War II by the US government. Their destructive power was extremely tremendous. In 1945, two bombs were dropped on two Japanese cities, *Hiroshima* and *Nagasaki*, causing complete destruction. The explosions killed about 200 000 people and caused many others to have physical and psychological illnesses.

Some countries claim that building nuclear weapons can help defend themselves if they are under attack. However, developing nuclear weapons do not bring peace to the world.

Ts may show Ss the following videos.

(1) Dropping an atomic bomb in Nagasaki

(2) The explosion of the first hydrogen bomb

(1) (2)



Historical photo showing the destruction of Hiroshima caused by the atomic bomb in the Second World War.

Exercise 3.3

1 Which of the following statements about nuclear waste (Fig a) is **incorrect**?



Fig a

A Nuclear waste is produced in nuclear power plants.

B Nuclear waste is radioactive.

C Special handling approaches for nuclear waste are needed.

D Nuclear waste would not release energy.

★**2** Which of the following are the advantages of generating electricity by controlled nuclear fusion over controlled nuclear fission?

(1) There is an abundant supply of fuel.

(2) The fusion products pose almost no harm to humans.

(3) It only requires an environment at a low temperature which is easier to handle.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

3 State two advantages and two disadvantages of using nuclear power.

atomic bomb 原子彈　hydrogen bomb 氫彈　Hiroshima 廣島　Nagasaki 長崎

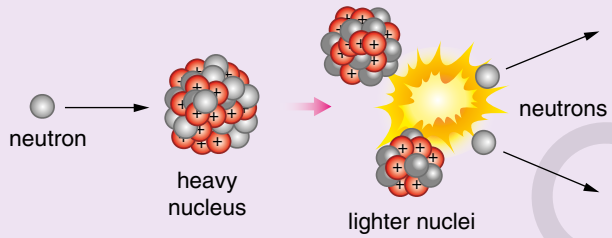
Q19 Exam Report: Part (a) was well answered. Some candidates were unable to identify the product *X* despite they had found the correct proton and mass numbers. In (b), most managed to find the mass defect correctly. Some failed to convert the mass defect to energy in MeV. Weaker ones mistook the energy equivalent of the mass of an α particle as the minimum kinetic energy or omitted the α particle or the proton in finding the minimum kinetic energy. Candidates' performance in (c) was poor. Only a few were able to give correct explanations based on the conservation of momentum. Popular explanations included 'extra energy is needed because of inelastic collision or binding energy', 'to compensate for the energy loss to surroundings' or 'to overcome repulsive forces'.

Key concepts 3

1 Nuclear fission and fusion

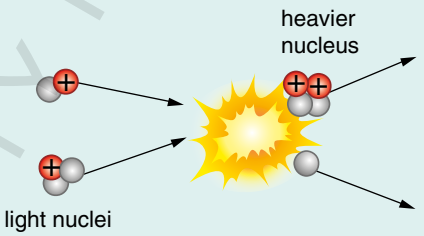
Nuclear fission

- A nucleus splits into lighter nuclei
- Heavy nucleus splits ⇒ releases energy



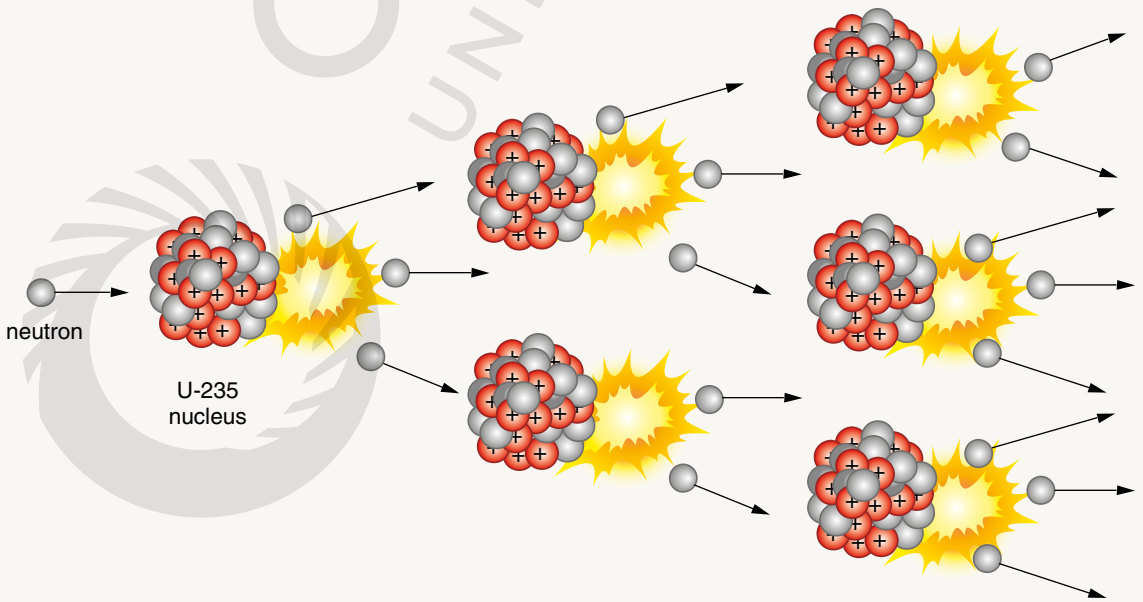
Nuclear fusion

- Two nuclei join together to form a heavier nucleus
- Light nuclei join ⇒ release energy
- Requires high temperatures (~10^7 K) to occur



2 Chain reaction

- Neutrons released in nuclear fission split other nuclei
- Is sustained if at least one neutron on average from each fission causes another fission



Critical mass

= minimum amount of fissile material required for a sustained chain reaction

Q15 Exam Report: Parts (a) and (b) were well answered. In (a), a few candidates wrote the equation with wrong notation. In (b), weaker candidates made mistakes in converting units like u, MeV and J. Candidates' performance in part (c) was fair. Many were weak in calculation with moles and they mixed up curie (Ci) and disintegrations per seconds (or Bq) when expressing the activity.

3 Mass-energy relationship Ext

- Nuclear reactions can release huge amounts of energy
- Mass is converted into energy in the process

total mass of particles before reaction = total mass of particles after reaction + energy released

Conversion between mass and energy in different units

$\Delta E = \Delta m c^2$

J kg 3.00×10^8 m s^{-1}

$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$

$1 \text{ u} = 1.661 \times 10^{-27} \text{ kg}$

1 u is equivalent to 931 MeV

4 Applications of nuclear energy

Controlled nuclear reactions

Using nuclear reactions to generate electricity



controlled nuclear fusion (under development)



controlled nuclear fission (currently in use)

Key terms

Ext **1** atomic mass unit (u) 原子質量單位 p.100
2 chain reaction 連鎖反應 p.93
3 controlled nuclear fission 受控核裂變 p.107
4 controlled nuclear fusion 受控核聚變 p.112
5 critical mass 臨界質量 p.94
Ext **6** electron-volt (eV) 電子伏 p.99
Ext **7** mass-energy relationship 質能關係 p.98
8 nuclear fission 核裂變 p.92
9 nuclear fusion 核聚變 p.92
10 nuclear reactor 核反應堆 p.107
11 nuclear waste 核廢料 p.108

Revision 3

Take 1 u = 1.661×10^{-27} kg, 1 u is equivalent to 931 MeV, $c = 3.00 \times 10^{8}$ m s^{-1}.

Multiple-choice questions

★1 3.1 Current nuclear power plants make use of nuclear fission to generate electricity. Which of the following statements about the fission reaction used is/are correct?

(1) It is a kind of spontaneous reaction.

(2) The electrostatic repulsion between nuclei must be overcome for this reaction to start.

(3) The mass of fission fuel should exceed a certain value to sustain a chain reaction.

A (3) only
B (1) and (2) only
C (1) and (3) only
D (2) and (3) only

★2 Ext 3.2 The mass of a water molecule is 2.99×10^{-26} kg. This mass is equivalent to an energy of

A 1.35×10^{-9} J.
B 5.38×10^{-9} J.
C 8410 MeV.
D 16 800 MeV.

★3 3.1 Which of the following conditions is/are necessary to sustain a chain reaction in the fission of uranium?

(1) The temperature is over 10^{7} K.

(2) More than one neutron on average is released in each fission reaction.

(3) All neutrons emitted in a fission reaction will take part in another fission reaction.

A (1) only
B (2) only
C (1) and (3) only
D (1), (2) and (3)

★4 3.1 Among the nuclear reactions represented by the following equations, which is/are nuclear fusion?

(1) ${}_{2}^{4}\mathrm{He} + {}_{2}^{4}\mathrm{He} \rightarrow {}_{4}^{8}\mathrm{Be}$

(2) ${}_{3}^{6}\mathrm{Li} + {}_{0}^{1}\mathrm{n} \rightarrow {}_{1}^{3}\mathrm{H} + {}_{2}^{4}\mathrm{He}$

(3) ${}_{3}^{6}\mathrm{Li} + {}_{1}^{2}\mathrm{H} \rightarrow {}_{4}^{7}\mathrm{Be} + {}_{0}^{1}\mathrm{n}$

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

★5 Ext 3.2 Arrange the following amounts of energy in ascending order.

(1) 1×10^{25} eV

(2) 1×10^{6} J

(3) 1 kW h

A (1), (3), (2)
B (1), (2), (3)
C (2), (1), (3)
D (2), (3), (1)

★6 Ext 3.2 A nuclear reaction is represented by the following equation:

$${}_{1}^{3}\mathrm{H} + {}_{1}^{3}\mathrm{H} \rightarrow {}_{2}^{4}\mathrm{He} + 2{}_{0}^{1}\mathrm{n}$$

The mass of a helium-4 nucleus is 4.001 505 u and that of a neutron is 1.008 665 u. If 1.82×10^{-12} J of energy is released during the reaction, which of the following gives the best estimation of the mass of a hydrogen-3 nucleus?

A 2.4990 u
B 2.5112 u
C 3.0033 u
D 3.0155 u

★7 Misc Uranium-235 only makes up 0.7% of the natural uranium. The concentration of uranium-235 is enriched to above 3% in the fuel used in nuclear power plants. Having a higher concentration of uranium-235,

A neutrons in the fuel travel at a higher speed.
B a neutron is more likely to trigger further fission when being released.
C more neutrons are produced when a uranium-235 nucleus undergoes fission.
D more energy is released when a uranium-235 nucleus undergoes fission.

8 HKDSE 2015 Paper 1A Q31 3.1

Which of the following nuclear reactions is/are **spontaneous** reaction(s)?

(1) ${}_{11}^{24}\mathrm{Na} \rightarrow {}_{12}^{24}\mathrm{Mg} + {}_{-1}^{0}\mathrm{e}$

(2) ${}_{5}^{10}\mathrm{B} + {}_{0}^{1}\mathrm{n} \rightarrow {}_{3}^{7}\mathrm{Li} + {}_{2}^{4}\mathrm{He}$

(3) ${}_{1}^{2}\mathrm{H} + {}_{1}^{3}\mathrm{H} \rightarrow {}_{2}^{4}\mathrm{He} + {}_{0}^{1}\mathrm{n}$

A (1) only (36%)
B (3) only (23%)
C (1) and (2) only (20%)
D (2) and (3) only (21%)

Conventional questions

***9** 3.1 In nuclear fusion, two hydrogen-2 nuclei are brought close to each other at high speeds (Fig a). A helium-3 nucleus is formed.



Fig a

(a) State the meaning of nuclear fusion. (1 mark)

(b) Write down an equation for the above fusion reaction. (2 marks)

(c) Explain why the hydrogen-2 nuclei should be brought close to each other at high speeds. (2 marks)

***10** Misc The mass of the Sun (Fig b) is 1.99×10^{30} kg. It radiates energy at a power of 3.90×10^{26} W.

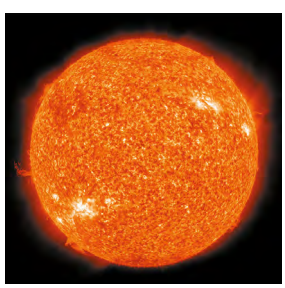


Fig b

(a) What kind of nuclear reaction is responsible for the large amount of energy radiated by the Sun? (1 mark)

Ext (b) About 0.07% of the mass of the Sun is converted into energy in a certain period of time. Find the length of the time period in years. Given: 1 year = 3.16×10^7 s. (3 marks)

***11** Ext Misc Assume that a nuclear fusion reactor is constructed. A series of nuclear reactions occur to generate energy. The net result of the reactions is described by the following equation:

$$3\,{}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{1}\mathrm{p} + {}^{1}_{0}\mathrm{n}$$

Given: mass of ${}^{2}_{1}\mathrm{H}$ = 2.013 553 u
mass of ${}^{4}_{2}\mathrm{He}$ = 4.001 505 u
mass of ${}^{1}_{1}\mathrm{p}$ = 1.007 276 u
mass of ${}^{1}_{0}\mathrm{n}$ = 1.008 665 u

(a) Find the total amount of energy (in J) released in the series of reactions. (3 marks)

(b) The amount of electrical energy consumed by Hong Kong in one day is about 4.5×10^{14} J. If this reactor were used as the only energy source for Hong Kong, estimate the number of hydrogen-2 atoms consumed every day. (2 marks)

***12** Misc In March 2011, a nuclear accident happened at the Fukushima Nuclear Power Station in Japan. Some radioactive substances leaked into the surrounding areas. One of the leaking substances was iodine-131. It is produced from the following reaction:

$${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{131}_{53}\mathrm{I} + {}^{102}_{39}\mathrm{Y} + x\,{}^{1}_{0}\mathrm{n}$$

(a) Find the value of x. (1 mark)

(b) Explain the importance of the neutron(s) produced in the reaction for a nuclear power station. (2 marks)

Ext (c) Given: mass of ${}^{235}_{92}\mathrm{U}$ = 234.993 422 u
mass of ${}^{131}_{53}\mathrm{I}$ = 130.877 029 u
mass of ${}^{102}_{39}\mathrm{Y}$ = 101.912 917 u
mass of ${}^{1}_{0}\mathrm{n}$ = 1.008 665 u

Find the energy released in the reaction in MeV. (2 marks)

****13** Ext Misc The following fission reaction occurs in a nuclear reactor:

$${}^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{140}_{54}\mathrm{Xe} + {}^{94}_{38}\mathrm{Sr} + 2\,{}^{1}_{0}\mathrm{n}$$

Given: mass of ${}^{235}_{92}\mathrm{U}$ = 234.993 422 u
mass of ${}^{140}_{54}\mathrm{Xe}$ = 139.892 009 u
mass of ${}^{94}_{38}\mathrm{Sr}$ = 93.894 494 u
mass of ${}^{1}_{0}\mathrm{n}$ = 1.008 665 u

(a) Find the energy (in J) released in this reaction. (3 marks)

(b) Describe how a chain reaction is formed in the reactor. (2 marks)

(c) If a power of 1200 MW is delivered by the nuclear fuels in the reactor and the uranium-235 nuclei only undergo the above reaction, find the number of uranium-235 nuclei consumed every second. (1 MW = 10^6 W) (2 marks)

Refer p.105

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14 Ext Misc A polonium-210 ($^{210}_{84}$Po) nucleus releases 5.4 MeV of energy by emitting an α particle and decays to a stable lead (Pb) nucleus. The half-life of polonium-210 is 138 days.

(a) The mass of an α particle is 4.0015 u. Estimate the speed of the α particle emitted in the decay. You may assume that all energy released in the decay becomes the kinetic energy of the α particle. (2 marks)

(b) There are 10^{10} polonium-210 nuclei in a sample.

(i) Find the activity of the sample. (3 marks)

(ii) Find the power (in W) delivered by the sample. (2 marks)

Refer p.105

See p.115 for the Exam Report.

15 Misc HKDSE 2012 Paper 1B Q11

Radium-226 ($^{226}_{88}$Ra) undergoes α decay into radon (Rn).

(a) Write a nuclear equation for the decay. (2 marks)

Ext (b) Given:

mass of a radium nucleus = 226.0254 u

mass of a radon nucleus = 222.0176 u

mass of an α particle = 4.0026 u

Calculate the energy released in the decay in MeV. (2 marks)

(c) 1 curie (Ci) is defined as the activity of 1 g of radium. The activity of a radium source used in laboratories is about 5 μCi. Estimate the number of radium atoms in this source and hence find its activity expressed in disintegrations per second. The half-life of radium-226 is 1600 years and take the mass of one mole of radium as 226 g. (1 μCi = 1×10^{-6} Ci) (3 marks)

See p.111 for the Exam Report.

16 Misc HKDSE 2014 Paper 1B Q10

Voyager I is a space probe designed by NASA to operate for over ten years in space. It was equipped with a radioisotope thermoelectric generator (RTG) which can convert the energy released from the decay of a radioactive source into electrical power. *Voyager I* operates with a plutonium-238 radioactive source that undergoes α decay.

(a) The plutonium-238 source is sealed inside a thin metallic casing of the RTG. The photo shows a NASA staff handling the RTG with his bare hands (Fig c). Explain why it is fine for it to be handled by the staff in this way. (1 mark)



Fig c

When *Voyager I* was launched, the number of plutonium-238 atoms in the source was 3.2×10^{25}.

Given: half-life of plutonium-238 = 87.74 years.

Take 1 year = 3.16×10^{7} s.

Ext (b) (i) Find the activity, in Bq, of the plutonium source at the time of launch. (3 marks)

(ii) When a plutonium-238 atom decays, it releases 5.5 MeV of energy. Estimate the power, in kW, delivered by the source at the time of launch. (2 marks)

(iii) The RTG of *Voyager I* is still in operation as *Voyager I* just left the solar system in September 2013 after it was launched 36 years ago. Estimate the corresponding power delivered by the plutonium source, expressed in **percentage** of the power delivered at the time of launch. (2 marks)

See p.111 for the Exam Report.

17 Misc HKDSE 2015 Paper 1B Q10

Scientists had been experimenting controlled fusion in a nuclear reactor in which deuterium ($^{2}_{1}$H) and tritium ($^{3}_{1}$H) undergo the following nuclear fusion:

$$^{2}_{1}\text{H} + {}^{3}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n} + \text{energy released}$$

Given:

mass of a deuterium nucleus = 2.014 102 u

mass of a tritium nucleus = 3.016 049 u

mass of a helium nucleus = 4.002 602 u

mass of a neutron = 1.008 665 u

Ext (a) Calculate the energy released, in MeV, in the above nuclear fusion. (2 marks)

(b) A deuterium nucleus and a tritium nucleus have to be within 10^{-15} m for nuclear fusion to occur and that a large amount of work done (about 0.4 MeV) is needed to bring two well separated nuclei to such a close distance.

(i) Explain why a large amount of work done is needed and state the kind of energy this work done has become. (2 marks)

In the nuclear reactor, deuterium and tritium exist as plasma, which is a mixture of ions at a very high temperature.

(ii) Explain why a very high temperature is needed for nuclear fusion to occur. (1 mark)

Ext (iii) Estimate the **order of magnitude** of the minimum temperature at which fusion of deuterium and tritium nuclei would be possible if the plasma can be regarded as an ideal gas. (Hint: For an ideal gas, the gas molecules each is assumed to have an average kinetic energy $E_K = \frac{3RT}{2N_A}$) (2 marks)

See p.111 for the Exam Report.

18 HKDSE 2019 Paper 1B Q10

Misc

(a) The equation below represents nuclear fission of uranium-235 (U-235).

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + x\,{}^{1}_{0}\text{n} + 200 \text{ MeV}$$

(i) What is the value of x? (1 mark)

(ii) State a necessary condition for chain reaction of fission to occur. (1 mark)

Scientists found evidence in Oklo, Africa that natural nuclear fission occurred two billion (2×10^9) years ago. The uranium mineral ore mined from Oklo **at present** is found to have 0.6% concentration by mass of U-235 (see Table a), which is much lower than usual.

(b) The table gives the information of U-235 and U-238 in a sample of uranium mineral ore found in Oklo.

Given: half-life of U-235 = 7.04×10^8 years

	2×10^9 years ago	at present
U-235	m_0 kg	0.060 kg (i.e. 0.6% concentration by mass)
U-238	13.556 kg	9.940 kg (i.e. 99.4% concentration by mass)

Table a

Ext (i) Estimate the amount m_0 (in kg) of U-235 in the sample 2×10^9 years ago. (2 marks)

(ii) Hence determine whether natural nuclear fission of U-235 was possible 2×10^9 years ago. For fission of U-235 to happen, its concentration by mass in the uranium mineral ore has to be at least 3%. (1 mark)

There must be underground water in the vicinity of this uranium-rich mineral deposit for natural nuclear fission to be possible. Since water can slow down the fast neutrons from fission, these neutrons can easily be captured by U-235.

(c) In fact the chain reaction stopped even before the concentration by mass of U-235 dropped to 3%. Explain why this occurred. (2 marks)

See p.114 for the Exam Report.

19 HKDSE 2020 Paper 1B Q10

Misc

Given: mass of proton = 1.0073 u

mass of α particle = 4.0015 u

mass of $^{14}_{7}\text{N}$ nucleus = 13.9993 u

mass of $^{17}_{8}\text{O}$ nucleus = 16.9947 u

When a stationary $^{14}_{7}\text{N}$ nucleus bombarded by an α particle, the following nuclear reaction can be triggered with products $^{17}_{8}\text{O}$ and X fly off:

$$\alpha + {}^{14}_{7}\text{N} \rightarrow {}^{17}_{8}\text{O} + X$$

(a) What is X? (1 mark)

Ext (b) Based on energy consideration, estimate the minimum kinetic energy, in MeV, of the α particle required for such a nuclear reaction to occur. (2 marks)

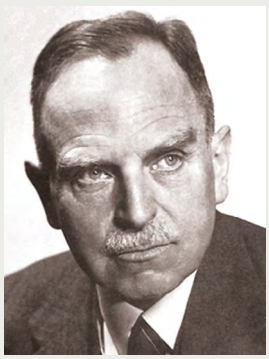
(c) However, when conservation of momentum is also taken into account, the α particle must possess a kinetic energy greater than that found in (b) to bring about such a reaction. Explain. (2 marks)

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Know more

Discovery of nuclear fission (p.92)

In 1938, *Otto Hahn* found that bombarding uranium with slow-moving neutrons could produce barium. *Lise Meitner* and *Otto Frisch* later showed that a uranium nucleus is unstable and can be torn apart by its own repulsive electrostatic force if disturbed by a neutron. This theory correctly deduced the energy released in the reaction and hence successfully explained the experimental result. Meitner and Frisch coined the term 'nuclear fission' in analogy to the binary fission of living cells. In 1944, Hahn was awarded the Nobel Prize in Chemistry for the discovery.



Otto Hahn (1879–1968).



Lise Meitner (1878–1968).



Otto Frisch (1904–1979).

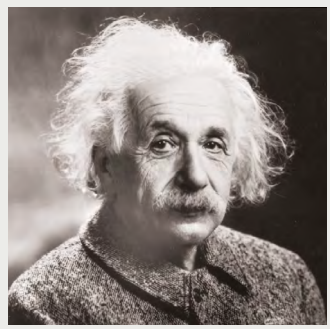
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Albert Einstein (1879-1955) (p.98)

Einstein made key contributions to many areas of modern physics. One of his most notable achievements was the establishment of the theory of relativity. At the age of 26, Einstein proposed the special theory of relativity, which replaced Newton's laws of motion for objects with high energy. In the next ten years, he expanded this theory and completed the *general theory of relativity*, which replaced Newton's law of gravity. This theory is now the foundation of *cosmology*.

In Book E2, we will see how he also changed our understanding of the nature of light.

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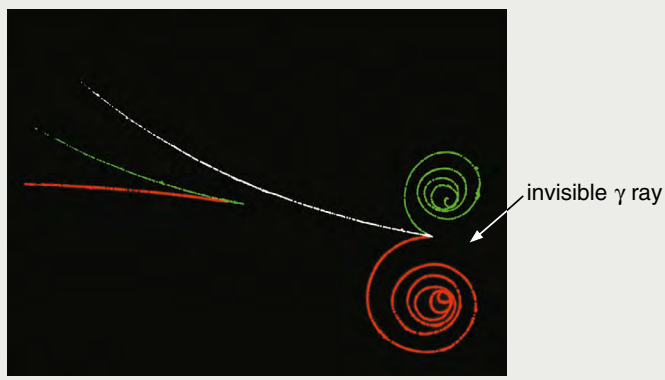


Albert Einstein.

Conversion between mass and energy (p.98)

According to the special theory of relativity, if an isolated stationary object emits electromagnetic waves, then its mass must decrease. This idea was confirmed by later experiments in particle physics. Furthermore, it was discovered that particles can be converted into electromagnetic waves, and electromagnetic waves can be converted into particles.

The photo is taken in a bubble chamber in a magnetic field, showing the formation of two pairs of electrons and positrons. The pair on the right move in spiral tracks and the pair on the left are more energetic and moves in less curved paths. The white line extending from the spiral tracks shows a path of an atomic electron which is displaced by the γ radiation.



γ rays are converted into electrons (paths shown in green) and positrons (paths shown in red).

Otto Hahn 哈恩　Lise Meitner 麥特勒　Otto Frisch 弗里施
general theory of relativity 廣義相對論　cosmology 宇宙學